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A PALAEOMAGNETIC STUDY OF THE CYGNET ALKALINE COMPLEX OF TASMANIA

By W. A. ROBERTSON AND L. HASTIE

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ABSTRACT

This paper describes a palaeomagnetic study of the Cygnet syenite boss, associated dykes, hybrid rocks, and baked sediments, giving new information relevant to the age of the intrusion. All these groups have parallel directions with a steep negative inclination, except those somewhat affected by a component induced by the earth's present field. The latter component has been minimized by treatment in alternating magnetic fields and the resulting directions, which are parallel to those in the Tasmanian dolerite, are interpreted as being the direction of the earth's field at the time of cooling. This suggests that the Cygnet intrusions are probably closely related in time to the Tasmanian dolerites, as Carey (1958) has argued from structural considerations.

INTRODUCTION

The age of the Cygnet Alkaline Complex is known only within wide limits and varying estimates of its most probable age have been put forward. Twelvetrees and other early workers (Twelvetrees and Petterd, 1898), tentatively placed the intrusion of the Cygnet Alkaline Complex in the Permo-Carboniferous epoch. Skeats (1917), mainly on the evidence of intrusion of dolerite by alkaline porphyry of the complex at Little Oyster Cove, concluded (p. 164) that the Cygnet Alkaline Complex "can be referred to the Kainozoic period with a high degree of probability". David (1950, p. 581) says, "A series of alkaline intrusions, now assigned a Tertiary age, occurs in the south of Tasmania at Port Cygnet".

A detailed petrological study by Edwards (1947) of the unusual rock types from the complex showed that they are due to hybridization between the dolerite and Cygnet alkaline material. The age limits for the intrusion of the dolerite are Upper Triassic to Lower Tertiary (Banks, 1958) so that they may be Jurassic or Cretaceous. By analogy with the Karroo dolerites of South Africa Jurassic is considered the more probable (Banks, 1958). Thus the Cygnet Alkaline Complex was probably intruded during or later than the Jurassic Period. Isostrat studies have shown that the dolerite in the Lower Huon area has the form of a flat cone sheet with its apex near Port Cygnet. The Cygnet Alkaline Complex is at the centre of this cone sheet, giving rise to the proposition that it was intruded through the same centre, and may thus be cognate with the dolerite (Carey, 1958).

A palaeomagnetic study of the Cygnet Alkaline Complex may be relevant to the problem of its age since the directions of magnetization can be compared with those from the Lower Tertiary Volcanics of Victoria (Irving and Green, 1957) and the Tasmanian dolerite (Irving, 1956) which are already known. This work was undertaken with this object in mind.

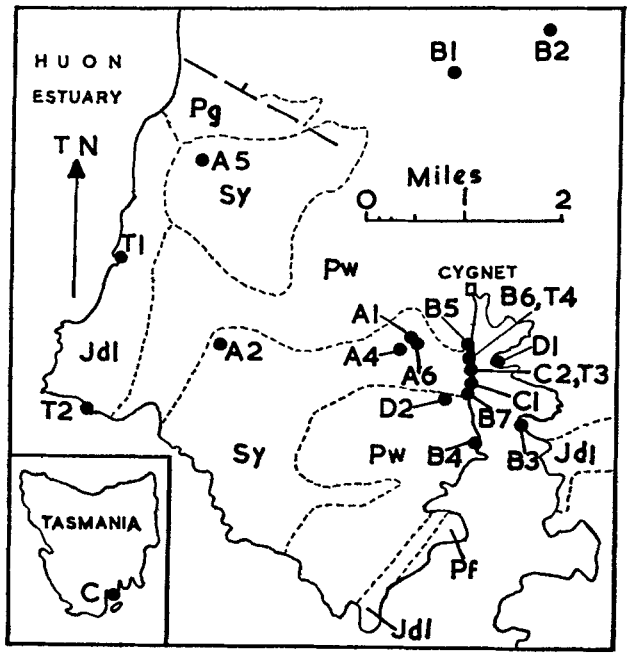


Fig. 1. Sketch map showing sampling sites. Sites *A*1-*A*6 are in the syenite boss, *A*3 is at Peppermint Bay, 8¼ miles east-north-east of Cygnet, *B*1-*B*7 in dykes, *C*1 and *C*2 from contacts with dolerite, and *D*1 and *D*2 baked sediment. Sites *T*1-*T*4 are in Tasmanian dolerite.

Sy = Alkali-syenite intrusive boss.
Jdl = Tasmanian dolerite.
Pf = Ferntree Mudstone
Pw = Woodbridge Glacial Formation } Permian (Smith, 1957)
Pg = Grange Mudstone
Map and Geology after R. J. Ford.

GEOLOGY AND SAMPLING

The alkali-syenite boss-like intrusion at Cygnet (43°9·9′S., 147°5·8′E.) is indifferently exposed over an area of about 25 square miles near the east coast of Tasmania, 30 miles south of Hobart (Fig. 1, Edwards, 1947). Associated with the syenite is a series of petrologically similar dykes (Edwards, 1947). The main mass of syenite intrudes Permian sediments and dolerite. The dykes intrude sediment, dolerite and syenite. The sediments are baked at the margin.

At the dolerite contact potash-rich hybrid rocks have been formed in a zone about 300 yards wide.

At least two oriented specimens have been taken from six sites in the boss (*A*1-*A*6), seven dykes (*B*1-*B*7), two sites in the hybrid zone (*C*1 and *C*2) and

TABLE 1

Site Localities

Site	Lat. S.	Long. E.	Collection Locality
A1	43°10.4′	147° 5.1′	Quarry in dark, marginal syenite, ¾ mile W.S.W. of Cygnet G.P.O.
A2	43°11.4′	147° 2.7′	Road cutting in porphyritic syenite, 2¾ miles W. by S. of Cygnet G.P.O.
A3	43° 8.9′	147°15.7′	Not used as weathered. Cliff at Peppermint Bay, 8½ miles E.N.E. of Cygnet G.P.O.
A4	43°10.6′	147° 4.8′	Blast hole in porphyritic syenite about 250 yards S.W. of Site A1.
A5	43° 9.1′	147° 2.1′	Blast hole in porphyritic syenite 3 miles W.N.W. of Cygnet G.P.O.
A6	43°10.4′	147° 5.1′	Blast hole in porphyritic syenite about 150 yards S.E. of A1.
B1	43° 7.9′	147° 4.8′	Cliff in porphyritic garnetiferous dyke 2¼ miles N. of Cygnet G.P.O.
B2	43° 7.7′	147° 6.8′	Cliff in haune-garnet-sanadine porphyry dyke, 8 ft. wide striking 175/60W. intruding Permian Woodbridge Mudstone, 2¾ miles N. by E. of Cygnet G.P.O.
B3	43°11.4′	147° 6.8′	Gave random directions and not used. Garnet-feldspar porphyry dyke on foreshore striking 205/steep N.W. intruding Permian Woodbridge sediments 1½ miles S. by E. of Cygnet G.P.O.
B4	43°11.6′	147° 6.5′	Dyke 2 ft. wide on foreshore containing whorls of sanadine phenocrysts. Striking 152/70E. and intruding Permian Woodbridge Mudstone. 1½ miles S. of Cygnet G.P.O.
B5	43°10.6′	147° 6.2′	Strongly fractured dyke of syenite porphyry on foreshore ⅝ mile S. of Cygnet G.P.O.
B6	43°10.9′	147° 6.2′	Syenite-porphyry dyke on foreshore, ¾ mile S. of Cygnet G.P.O.
B7	43°11.1′	147° 6.2′	Aplitic dyke striking 075/steep N. on foreshore 1 mile S. of Cygnet G.P.O.
C1	43°10.9′	147° 6.2′	Contact between feldspar pegmatite and hybridized dolerite from zone of hybridization on foreshore 1 mile S. of Cygnet G.P.O.
C2	43°10.9′	147° 6.2′	From zone of hybridization of dolerite by syenite magma on foreshore 1¼ miles S. of Cygnet G.P.O.
D1	43°10.9′	147° 6.8′	Baked, metasomatized fine-grained blue-grey Permian sediments on foreshore ⅞ mile S. by E. of Cygnet G.P.O.
D2	43°11.4′	147° 5.5′	Baked sediment from roadside quarry 1¼ miles S. by W. of Cygnet G.P.O.
T1	43°10.4′	147° 0.7′	Dolerite from road cutting 3½ miles W. by N. of Cygnet G.P.O.
T2	43°11.9′	147° 0.7′	Coarse-grained dolerite from point on foreshore 4 miles W. by S. from Cygnet G.P.O.
T3	43°10.9′	147° 6.2′	Thermally altered dolerite from shoreline ¾ mile S. of Cygnet G.P.O.
T4	43°10.9′	147° 6.2′	Thermally altered dolerite from shoreline 150 yds. N. of T3.

two in baked sediment (*D*1 and *D*2), and also four from the dolerite (*T*1-*T*4). The localities are shown in Fig. 1 and listed in Table 1.

Most natural exposures are heavily weathered and it was often necessary to blast small holes with dynamite to obtain fresh rock. Samples have been obtained either from these or from quarries or road-cuttings, or along the shore. The type of exposures are detailed in Table 1. Two or more cylindrical specimens (35 mm. diameter x 35 mm. height) were cut from each sample. Altogether 59 samples from 21 sites have been studied.

DIRECTIONS OF MAGNETIZATION

The directions of natural remanent magnetization (NRM) observed in all specimens are plotted in Figs. 2a (boss), 2c (dykes) and 2e (baked and hybridized contacts), and the mean site directions are listed in Table 2. The individual site directions are significantly different from the directions of both the present (**P**) and the dipole (**D**) fields, but there is a tendency for directions in some specimens to be close to (**P**) and (**D**). This is interpreted as due to a small component imposed by the present earth's field. Although it is clear from Fig. 2 that this effect is small, it was thought worthwhile to attempt to remove this component by partial demagnetization in alternating magnetic fields. This is possible since the stability of these components is less than that of the original magnetization and so may be preferentially removed.

All specimens have been treated in an alternating field of 75 oersteds peak value using the apparatus previously described (Irving, Stott and Ward, 1961). The agreement of directions within each site was thus improved (see Fig. 2 and Table 1). In addition, this treatment changed the mean direction of some sites away from the present field; in the others the direction changes were not significant. After this treatment the mean site directions from the boss, dykes, baked and hybridized contacts are all approximately parallel to one another. Several specimens are also treated in higher alternating fields and demagnetization curves are given in Fig. 3. The directions do not change appreciably in fields up to 300 oersteds, but in higher fields they begin to scatter.

Three weathered samples from Site *A*3, at Peppermint Bay about eight miles east of Cygnet, were measured to test the effect of weathering. They gave directions close to the present field (**P**) which were not much changed by treatment in alternating fields. This suggests that the effect of weathering is to produce a very stable magnetization, presumably due to hematite, along the earth's present field. Two samples from Site *B*6, a syenite-porphyry dyke on the foreshore gave random directions ($R = 2{\cdot}4$, $N = 6$).

Three samples from site *C*1, in the hybrid zone, were taken from a contact between syenite-pegmatite and altered dolerite. Six specimens from the syenite and the dolerite side of the contact, and also from the contact itself, were measured. The mean directions from the three groups were not significantly different, although the scatter of directions increased from the dolerite to the syenite side; the standard deviation for the syenite is 20°, for the contact 17°, and 12° for the dolerite.

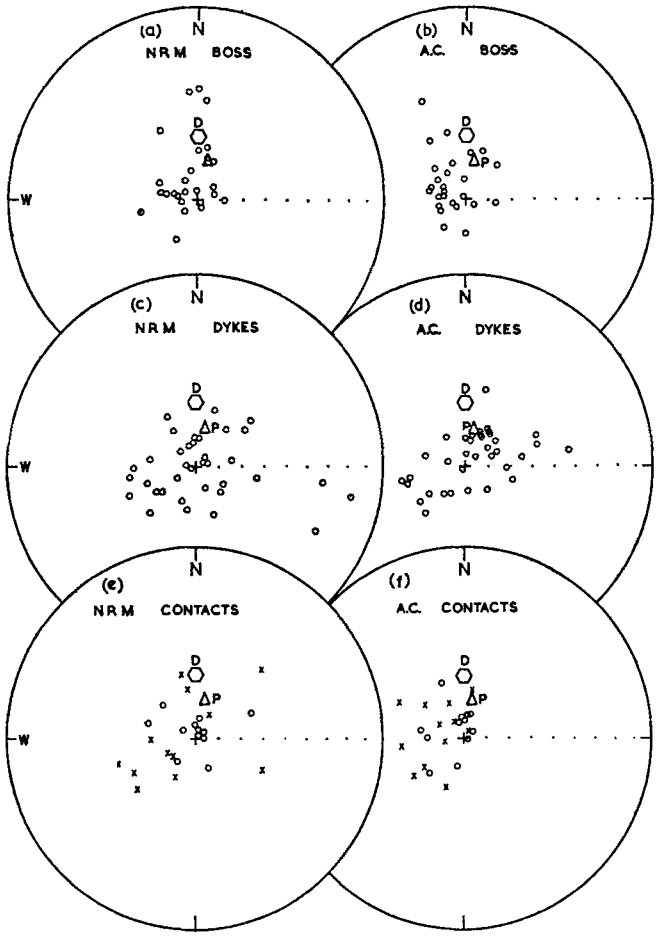


Fig. 2. Direction of magnetization. An equal-area Schmidt net is used, the plane of projection is horizontal, and open circles and crosses are north-seeking directions in the upper hemisphere (normal). The present (P) and dipole (D) field directions at Cygnet are given. 2(a) shows the NRM specimen directions from 5 sites in the syenite boss and 2(b) directions from the same specimens after treatment in an alternating peak field of 75 oersteds. 2(c) and 2(d) show respectively the NRM and treated specimen directions from 6 dykes. In 2(e) and 2(f), which show respectively NRM and treated specimen directions, the crosses represent directions from specimens from the two sites of Tasmanian dolerite heated by the boss, and the open circles represent directions from specimens from two sites of Permian sediment baked by the boss.

ANALYSIS

The directions used for analysis are those given in Figs. 2b, 2d and 2f, after treatment in 75 oersteds. The results from the weathered samples (site *A3*) are not included, nor are those from site *B6*. The mean direction at each site, with its 95 per cent. cone of confidence, a, and the standard deviation, s, were calculated using the method of Fisher (1953). The mean direction for each group is given in Table 3. The method of Watson and Irving (1957) gave an estimate of the within site (w) and between site (b) precision and the 95 per cent. cone of confidence. It is clear that there is no important difference in the directions between any of these groups. The overall mean obtained by giving each site unit weight for the Cygnet Alkaline Complex associated hybrids and contacts is also shown, together with that already obtained for the Tasmanian dolerite and the Older Volcanics of Victoria.

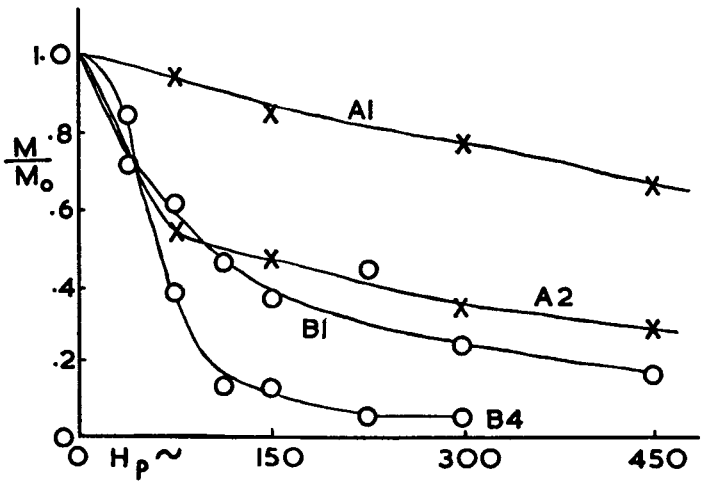


Fig. 3. Alternating field demagnetization curves. The symbols refer to the sites from which the specimens were taken. B6 yielded random directions, and is unstable. The ordinate gives the ratio of the intensity of magnetization remaining after treatment in the peak alternating field in oersteds recorded along the abscissa, to the NRM.

DISCUSSION

(a) *Origin of the magnetization.* The following facts suggest that the magnetization is thermoremanent (TRM) acquired at the time of intrusion.

(i) The directions, after minimizing secondary components, are different from those of the present and dipole fields.

(ii) The directions do not change significantly after treatment in alternating peak fields up to 450 oersteds, or in the case of the dykes, 300 oersteds, and

TABLE 2

Site mean directions

The initial directions, and those after treatment in an alternating field of peak value 75 oersteds, are given. S and N are the number of samples and specimens measured. D and I are the declination and inclination of the resultant site means, and R the length of the resultant vector. θ_0 is the standard deviation, in degrees, within which 63 per cent. of the measurements lie (Creer, Irving and Nairn, 1959), and a the half-angle of the 95 per cent. cone of confidence giving each specimen unit weight. M_0 and M_{75} are the intensities of magnetization before and after treatment.

				Initial directions					Directions after treatment					Intensity	
Group	Site	S	N	D	I	R	θ_0	a	D	I	R	θ_0	a	$X10^{-3}emu/cc$	
														M_0	M_{75}
A	A1	3	6	280	−78	5·91	10·7	9	291	−77	5·97	6·1	5	0·12	0·09
Syenite	A2	3	6	357	−55	5·85	14·1	12	338	−65	5·86	13·3	11	1·59	0·83
Boss	A4	2	6	77	−87	5·87	13·1	11	246	−87	5·91	11·0	9	0·02	0·02
	A5	1	2	334	−83	—	—	—	275	−87	—	—	—	0·39	0·50
	A6	3	6	321	−84	5·91	10·6	9	304	−84	5·85	14·3	12	0·52	0·40
B	B1	5	6	49	−77	5·74	18·7	16	70	−80	5·79	16·5	14	0·21	0·12
Dykes	B2	3	6	58	−81	5·77	17·7	15	58	−79	5·87	13·3	11	0·31	0·30
	B3	2	6	104	−58	5·03	36·2	32	88	−72	5·69	20·0	17	0·33	0·17
	B4	3	6	354	−85	5·98	5·7	5	33	−78	5·97	6·3	5	0·02	0·01
	B5	3	6	256	−88	5·71	19·6	17	345	−87	5·84	14·3	12	0·93	0·80
	B7	3	6	250	−65	5·95	8·1	7	250	−66	5·93	9·4	8	0·01	0·01
C	C1	3	6	222	−78	5·52	25·6	22	335	−68	5·90	11·5	10	0·82	0·21
Contract Dolerite-Syenite Hybrids	C2	4	6	325	−80	5·43	27·0	24	291	−72	8·5	19·1	13	0·41	0·20
D	D1	5	6	28	−85	5·81	15·9	13	301	−84	5·81	15·9	13	0·90	0·36
Contact Sediments baked by Boss	D2	2	6	301	−87	5·91	10·9	9	345	−84	5·93	9·4	8	0·02	0·01
T	T1	3	6	346	−83	5·94	9·1	—	—	—	—	—	—	0·43	—
Tasmanian dolerite near Intrusion	T2	3	6	344	−85	5·98	5·4	—	—	—	—	—	—	0·84	—
	T3	3	6	47	−88	5·94	8·9	—	—	—	—	—	—	4·51	—
	T4	2	6	208	−82	5·84	14·5	—	—	—	—	—	—	11·6	—

the demagnetization curves (Fig. 3), for Group A, are typical of those associated with TRM.

(iii) Groups C and D (Tables 2 and 3) obtained from hybridized dolerite and sediment respectively, heated by the intrusion of the boss, give directions parallel to those in the boss and the dykes. The intensity of magnetization of the sediment baked by the boss is at least 10 times stronger than that of similar sediment (Stott, private communication) unaffected by it.

(iv) To test the effect of heating on the dolerite 4 sites (*T*1-*T*4) were sampled; *T*1 and *T*2 from more than half a mile from the boss, and *T*3 and *T*4 within 150 yards of the hybrid zone. In the latter case the intensity was greater by a factor of 10 (Table 2).

TABLE 3

Mean directions of groups

The directions are those obtained after the specimens have been treated in an alternating peak field of 75 oersteds, giving each site unit weight. *B* is the number of sites. *D* and *I* are the mean declination and inclination for each group. *R* is the length of the resultant vector. *w* and *b* are the within site and between site dispersions respectively. a^1 is the half-angle of the 95 per cent. cone of confidence using the two-tier analysis of Watson and Irving (1957).

Group	*B*	*D*	*I*	*R*	*w*	*b*	a^1
A. Boss	5	313	−81	4·93	49	51	10
B. Dykes	6	57	−86	5·83	33	104	7
C. Dolerite hybrids	2	276	−75	—	—	—	—
D. Sediment contacts	2	327	−84	—	—	—	—
Average (ABC and D)	15	314	−85	14·64	56	776	5
Average Tasmanian dolerite (Irving 1956)	51	319	−84	50·05	41	111	—
Older Volcanics of Victoria (Irving *et al* 1957)	15	17	−73	—	—	35	—

(b) *Age of the Cygnet Intrusives.* The mean NRM direction of the Cygnet Alkaline Complex is very close to that of the dolerites of Tasmania. However, it is 18 degrees different from the direction of the Lower Tertiary Older Basalts of Victoria. Thus the results are consistent with Carey's argument from structural relations (Carey, 1958, pp. 1048-152), that the intrusion of the Cygnet Alkaline Complex was penecontemporaneous with that of the dolerite in Tasmania. This is contrary to previous interpretations on petrological evidence, which place it as possibly Tertiary (David, 1950, p. 581).

The pole position for the Cygnet result, assuming the earth's field to be a dipole (Creer *et al.*, 1957) (South pole 50S, 158E) is plotted in Fig. 4. The semi-axes ($d\Psi$, $d\chi$) along, and perpendicular respectively to the palaeomagnetic meridian for the area within which the true pole lies with a 95 per cent. proba-

bility are ($d\Psi = 12$, $d\chi = 12$) shown on Fig. 4. This makes clear the age relationships suggested above, indicating the Cygnet Alkaline Complex to be older than the Lower Tertiary Lavas of Victoria and comparable in age to the intrusion of the Tasmanian dolerites.

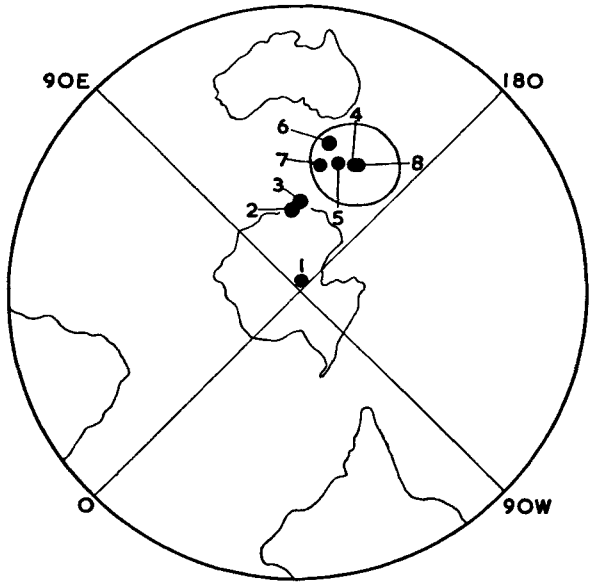


Fig. 4. Pole positions from Mesozoic and Kainozoic rocks in Australia. The poles are plotted in the southern hemisphere on an equal area net. 1, Newer Basalts of Victoria; 2, Older Basalts of Victoria; 3, Tertiary Basalts of New South Wales; 4, Tasmanian dolerite; 5, Prospect Dolerite; 6, Gibraltar Syenite; 7, Gingenbullen Dolerite; 8, Pole from Cygnet results, showing the oval within which the pole lay with a probability of 95 per cent. The close agreement between the Cygnet Alkaline Complex pole with that for the Tasmanian dolerite and also other intrusives from Eastern Australia (Boesen, Irving and Robertson, 1961) is noteworthy.

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REFERENCES

BANKS, M. R., 1958: A Comparison of Jurassic and Tertiary Trends in Tasmania. Dolerite, a Symposium, *Univ. of Tasm.*, p. 231.

BOESEN, R., IRVING, E., and ROBERTSON, W. A., 1961: The Palaeomagnetism of some Igneous Rock Bodies in New South Wales. *J. roy. Soc. N.S.W., 94,* pp. 227-232.

CAREY, S. W., 1958: The Isostrat, a New Technique for the Analysis of the Structure of the Tasmanian Dolerite. Dolerite, a Symposium, *Univ. of Tasm.*, p. 130.

CREER, K. M., IRVING, E., and NAIRN, A. E. M., 1959: Palaeomagnetism of the Great Whin Sill. *Geophys. J. roy. Astr. Soc., 2,* 4.

——, IRVING, E., and RUNCORN, S. K., 1957: Geophysical Interpretation of Palaeomagnetic Directions from Great Britain. *Phil. Trans. roy. Soc. Lond. (A), 250,* pp. 144-156.

DAVID, T. W. E., ed. W. R. BROWNE, 1950: *The Geology of the Commonwealth of Australia,* London, Arnold, p. 581.

EDWARDS, A. B., 1947: Alkali Hybrid Rocks of Port Cygnet, Tasmania. *Proc. roy. Soc. Vic., 58,* pp. 81-115.

FISHER, R. A., 1953: Dispersion on a Sphere. *Proc. roy. Soc. Lond. (A), 217,* pp. 295-305.

IRVING, E., 1956: The Magnetization of the Mesozoic Dolerites of Tasmania. *Proc. roy. Soc. Tasm., 90,* pp. 157-168.

——, and GREEN, R., 1957: The Palaeomagnetism of the Kainozoic Basalts of Victoria. *Mon. Not. R. astr. Soc. geophys. Suppl., 7,* 6, pp. 347-359.

——, STOTT, P. M., and WARD, M. A., 1961: Demagnetization of Igneous Rocks by Alternating Magnetic Fields. *Phil. Mag., 6, 62,* pp. 225-241.

SKEATS, E. W., 1917: On the Age of the Alkali Rocks of Port Cygnet and the D'Entrecasteaux Channel in S.E. Tasmania. *Proc. roy. Soc. Vic., 29,* pp. 154-164.

SMITH, E. M., 1957: Lexicon of the Stratigraphy of Tasmania. *Bur. Min. Resour. Aust.,* Canberra.

TWELVETREES, W. H., and PETTERD, W. F., 1898: On Hauyne Trachyte and Allied Rocks in the Districts of Port Cygnet and Oyster Cove. *Proc. roy. Soc. Tasm.*, 1898-99.

WATSON, G. S., and IRVING, E., 1957: Statistical Methods in Rock Magnetism. *Mon. Not. R. astr. Soc. geophys. Suppl., 7,* 6, pp. 290-300.

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